

Amortising the Integrality Gap: Imitation Learning for Real-Time Charging-and-Break Scheduling of Electric Trucks

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Abstract

Battery-electric trucks must schedule charging jointly with the rest and break periods that driving-hours regulation imposes, under stochastic travel times. A rolling-horizon look-ahead that solves its tail exactly is the strongest implementable policy we have for this problem, but it costs 45–90 s per decision and up to four hours of compute per route: a *batch* policy, not an operational one. We ask how much of it survives amortisation. Cloning it into a 36k-parameter masked network yields a policy that decides in **0.74 ms** — roughly $10^5\times$ faster — and lands $+0.43 \pm 0.18\%$ of realised route duration behind its own teacher on held-out instances, against $+2.74\%$ for a rule-based heuristic of comparable speed. The route there is not automatic: our design decisions were repeatedly made against *closed-loop* evidence that contradicted the supervised metrics, and the change that mattered most was a class-weighting choice that raised balanced accuracy from 0.299 to 0.477 while making realised duration six times worse. We also report a negative result: DAGger improved the development set but did not transfer to the held-out set. We argue that in sequential operational settings, offline predictive accuracy is not merely a weak proxy for policy quality but is frequently *anti-correlated* with it.

1 Problem and motivation

A battery-electric heavy truck on a long corridor faces two coupled schedules. Charging is dictated by energy and by where stations are; rest and break periods are dictated by EU driving-hours regulation [4], on a corridor whose station spacing is itself set by regulation [5]. The two schedules interact: a mandatory break taken *inside* a charging dwell is free, because the two run in parallel, so *when* to charge is inseparable from *when* to rest. Travel times are stochastic, so the schedule must be decided online, stop by stop.

Our companion study benchmarks a family of policies on this problem — a myopic rule-based heuristic, static robust plans, a two-stage stochastic plan with recourse, a rolling-horizon look-ahead, and a per-realisation hindsight oracle. Here we take that benchmark as given and ask a narrower question about the best implementable member of it.

Where the remaining headroom is. A one-at-a-time sweep of the look-ahead’s parameters is unusually uninformative in two of its three dimensions. Raising the scenario count from 10 to 50 moves realised duration by at most 0.4%; extending the horizon beyond one duty cycle (24 h) moves it by at most 0.2%, although *shortening* it below one duty cycle costs up to 1.7%. The one live axis is **integrality**: solving the look-ahead tail as a MIP rather than an LP relaxation roughly halves the gap to the hindsight optimum (Table 1), and costs an order of magnitude more time per decision.

The gap this paper attacks. Exactness is therefore worth about two points of optimality and is now our standard configuration — but at 45–90 s per decision (median, by route class) and 0.6–4.1 h of wall clock per route it is affordable only as an offline campaign. It cannot be re-evaluated continuously in a cab, it cannot be run across a fleet by a dispatcher, and it requires a commercial solver licence at the point of use. We therefore ask: *how much of an exact look-ahead survives being compressed into a policy that decides in milliseconds?* In Powell’s taxonomy [8] this converts a direct look-ahead into a policy-function approximation; in the ML-for-combinatorial-optimisation framing [2] it is amortised optimisation [3, 7]. Unlike neural construction heuristics for routing [6] the sequence of stops here is fixed and the learned object is the operational schedule; the closest application is safe reinforcement learning for dynamic electric vehicle routing [1].

Table 1: The look-ahead’s live axis. Solving the tail exactly halves the gap to the hindsight optimum and costs $\sim 10\times$ the decision time. Gap is the median distance to the per-realisation oracle; t_{dec} is the median decision time per stop. Reproduced from the companion benchmark.

Tail	Windows	Short routes		Medium routes	
		Gap (%)	t_{dec} (s)	Gap (%)	t_{dec} (s)
LP relaxation	none	4.3	4	4.6	18
LP relaxation	tight	7.6	4	6.8	18
Exact (MIP)	none	2.1	48	2.4	84
Exact (MIP)	tight	2.9	61	2.2	90

Table 2: Held-out test set (108 instances, evaluated once). Duration is the median paired difference from the exact-tail teacher over three training seeds; lower is better and 0 is teacher parity. Halts are routes ended by a regulatory breach. The student reaches teacher-comparable quality at $\sim 10^5\times$ less online compute.

Policy	Duration vs. teacher (%)	Halts /108	TW misses	t_{dec}	Solver
Greedy heuristic	+2.74	0	166	0.03 ms	no
Student (cloned)	+0.43 $\pm$ 0.18	2.7 $\pm$ 0.5	128	0.74 ms	no
Student + DAGger	+0.48 $\pm$ 0.31	4.7 $\pm$ 0.9	126	0.79 ms	no
Exact-tail look-ahead (teacher)	— (reference)	0	53	70,446 ms	yes

2 Method

Decision process. At stop i the state comprises the vehicle’s state of charge, the three regulatory clocks (consecutive driving, shift driving, shift work), the split-break flag, the reduced-rest and extended-shift budgets consumed, and the arrival time; plus a description of the road ahead — the next K nodes (leg duration and energy, whether the node is a charger or a customer, queue delay, and time-window slack *relative to now*) and aggregates over the whole remaining route. With $K = 20$ this is 141 features. The action is the joint discrete choice $a_i \in \{\text{charge}, -\text{charge}\} \times \{\text{none}, b_{15}, b_{30}, b_{45}\} \times \{\text{none}, r_1, r_2\}$, of which 12 combinations occur in the data, together with a charge duration τ_c . The reward is the negative increment of the benchmark’s own objective, so training and evaluation optimise the same quantity.

Policy. A two-layer MLP ($141 \rightarrow 128 \rightarrow 128$; 36,365 parameters) with two heads: masked logits over the 12 actions, and a softplus charge duration. Two constraint layers wrap it, both reusing the benchmark’s existing feasibility checks, so the learned policy is subject to exactly the rules every other policy obeys. *Masking* sets structurally impossible actions to $-\infty$ before the softmax (no charging away from a station; no reduced rest without budget). *Forcing* restricts the choice set further when a one-step check declares a charge, break or rest mandatory — the network still chooses freely *among* compliant actions, so the rules constrain rather than decide. Charge duration is clamped below by the energy needed to reach the next station at the 0.5-quantile of consumption, the same rule the teacher uses to size its own committed charge. No solver runs at deployment.

Data. 767 exact-tail look-ahead runs over a $3 \times 3 \times 4$ grid of route length, customer density and time-window class supply **74,380** labelled decisions; the simulator records the per-stop state, so no replay is needed. Instances are split by seed into 530 train / 129 validation / 108 test. Every design decision below was made on validation; the test set was evaluated once.

3 Experiments

Every policy is executed in the same simulator, over the same recorded travel-time realisations, and scored by the same metrics. A run that breaches a regulatory limit *halts* at that stop, so a route duration exists only for a completed route: halted runs are counted, never averaged into a duration. We report the paired difference in realised duration against the exact-tail teacher on the same instance, over three independently trained seeds.

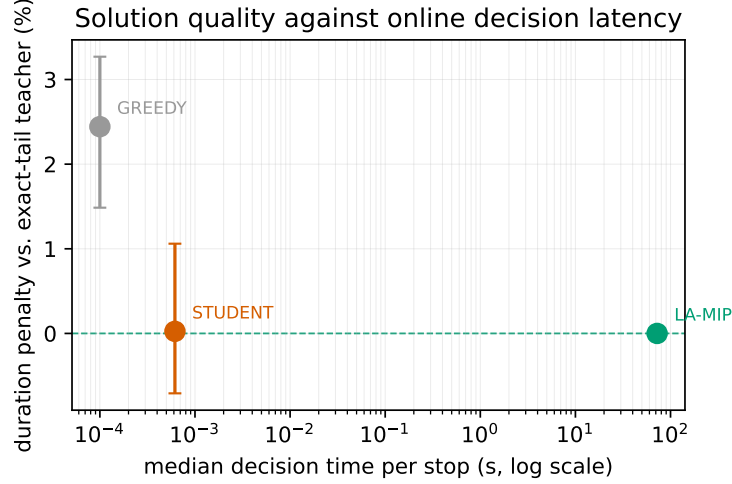


Figure 1: Quality against online decision latency (log scale), validation set. The student attains the horizontal of the exact-tail look-ahead at the latency of the myopic heuristic. Bars are interquartile ranges over instances.

Table 3: Design decisions, validation set. Balanced accuracy is mean per-class recall on held-out decisions; duration is the median paired difference from the teacher. In both rows the offline metric prefers the worse policy.

Decision	Setting	Balanced acc. $\uparrow$	Duration vs. teacher $\downarrow$
Class weighting	inverse-frequency ($p=0.5$)	0.477	+3.17%
	unweighted ($p=0$)	0.299	+0.56%
Look-ahead features	$K = 40$ (261 features)	0.452	+2.86%
	$K = 20$ (141 features)	0.299	+0.56%

Main result. Table 2 and Figure 1 give the headline. The cloned policy sits 0.43% behind a teacher that takes 70 s per decision, while deciding in 0.74 ms and requiring no solver; the rule-based heuristic, which is equally fast, is six times further behind. The student halts on 2.7 of 108 routes against the teacher’s zero — the residual cost of amortisation. Those failures are not a diffuse loss of quality: every one of them is a breach of the 15 h daily-spread ceiling.

Behavioural fidelity. Aggregate numbers hide whether the policy has learned the *structure* of the problem or merely its cost. Figure 2 shows one realised schedule against the teacher’s: rest placement, charging episodes and the state-of-charge trajectory are essentially superimposed, and both policies hide almost every mandatory break inside a charging dwell — the coupling that makes this problem non-separable. Quantitatively the student charges for 6.74 h per route against the teacher’s 6.80 h (−0.9%) and runs the pack to the same depth (5th-percentile state of charge 29.5% against 28.2%): it has learned that charge is time, not merely that empty batteries are bad.

Ablations, decided in closed loop. Table 3 pairs each design decision’s supervised metric with its realised effect. The pattern is consistent, and is the most useful thing we have to report: *the supervised metric moves the wrong way*. Inverse-frequency class weighting is the clearest case. The decision distribution is dominated by “keep driving” (87.9%), so weighting rare classes up is the textbook response, and by its own measure it works — balanced accuracy rises from 0.299 to 0.477. But it places the majority class at weight 0.035 against 5.3 for the rarest, so predicting a *rest* when the truth is *drive on* becomes nearly free in the loss, while costing 9–11 h of route time in reality. Realised duration degrades from +0.56% to +3.17%. The loss had quietly decorrelated from the objective, and no offline metric revealed it.

A negative result: DAGger did not transfer. Behaviour cloning trains on the teacher’s state distribution, and a policy that acts creates its own; DAGger [9] is the standard remedy. We ran one round: the student drove 530 training instances while the exact-tail teacher labelled a sampled 35% of the states it reached, yielding **17,941** aggregated labels (19.4%

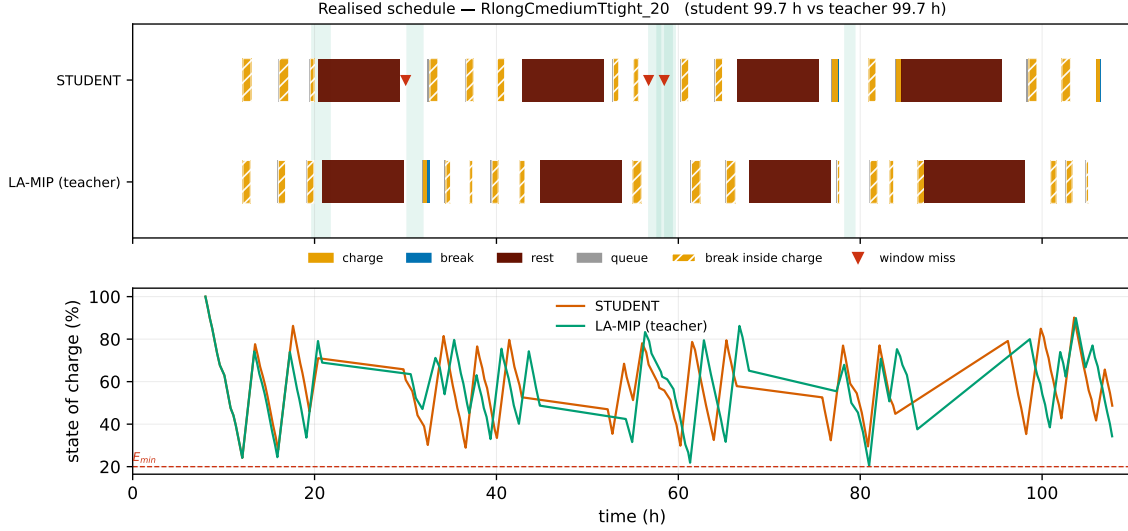


Figure 2: Realised schedule, student (top lane) against exact-tail teacher (bottom lane), on a held-out long route with tight windows. Hatched charging bars are dwells that absorb a mandatory break at no extra cost. Lower panel: state of charge.

of the training set). The distribution shift is real and measurable — at student-visited states the teacher prescribes rest markedly more often than in the base data (r_1 : 0.33% $\rightarrow$ 0.81%; r_2 : 1.63% $\rightarrow$ 2.33%), consistent with a policy that arrives at stops with its clocks closer to their limits. On validation the aggregated model looked better (median +0.19% against +0.44%; halts 3.3 against 5.7). *On the held-out set the effect reversed* (Table 2): +0.48% against +0.43%, with more halts rather than fewer. The validation gain was within twice the seed-to-seed spread and did not survive. We suspect a single DAgger round is the wrong tool once the state distribution has already been repaired by fixing the loss.

4 Discussion

An exact-tail look-ahead can be compressed into a 36k-parameter network that decides $\sim 10^5\times$ faster and gives up about half a percent of route duration. That is the difference between a policy that can be run as an offline study and one that can be run in a vehicle.

The transferable lesson is methodological. Repeatedly an offline metric and a closed-loop measurement disagreed, and each time the closed-loop measurement was the one worth believing; with class weighting, following standard practice would have made the deployed policy six times worse. In sequential operational problems the cost of an error is wildly non-uniform across the action space — a misplaced daily rest costs nine hours, a misplaced charging decision costs minutes — and any offline loss that treats them comparably has stopped measuring what matters. We would encourage reporting realised closed-loop cost as the primary metric for learned policies in operations, with predictive accuracy relegated to a diagnostic.

Limitations. Single vehicle; synthetic travel-time uncertainty from a fitted shifted lognormal; imitation bounds the policy by its teacher by construction; three training seeds. The residual failures concentrate in one regulatory constraint (the daily spread ceiling), which suggests a targeted feasibility layer rather than more data. Extending to a fleet sharing charging infrastructure is the natural next step, and turns a single-agent problem into a multi-agent one.

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